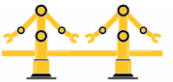
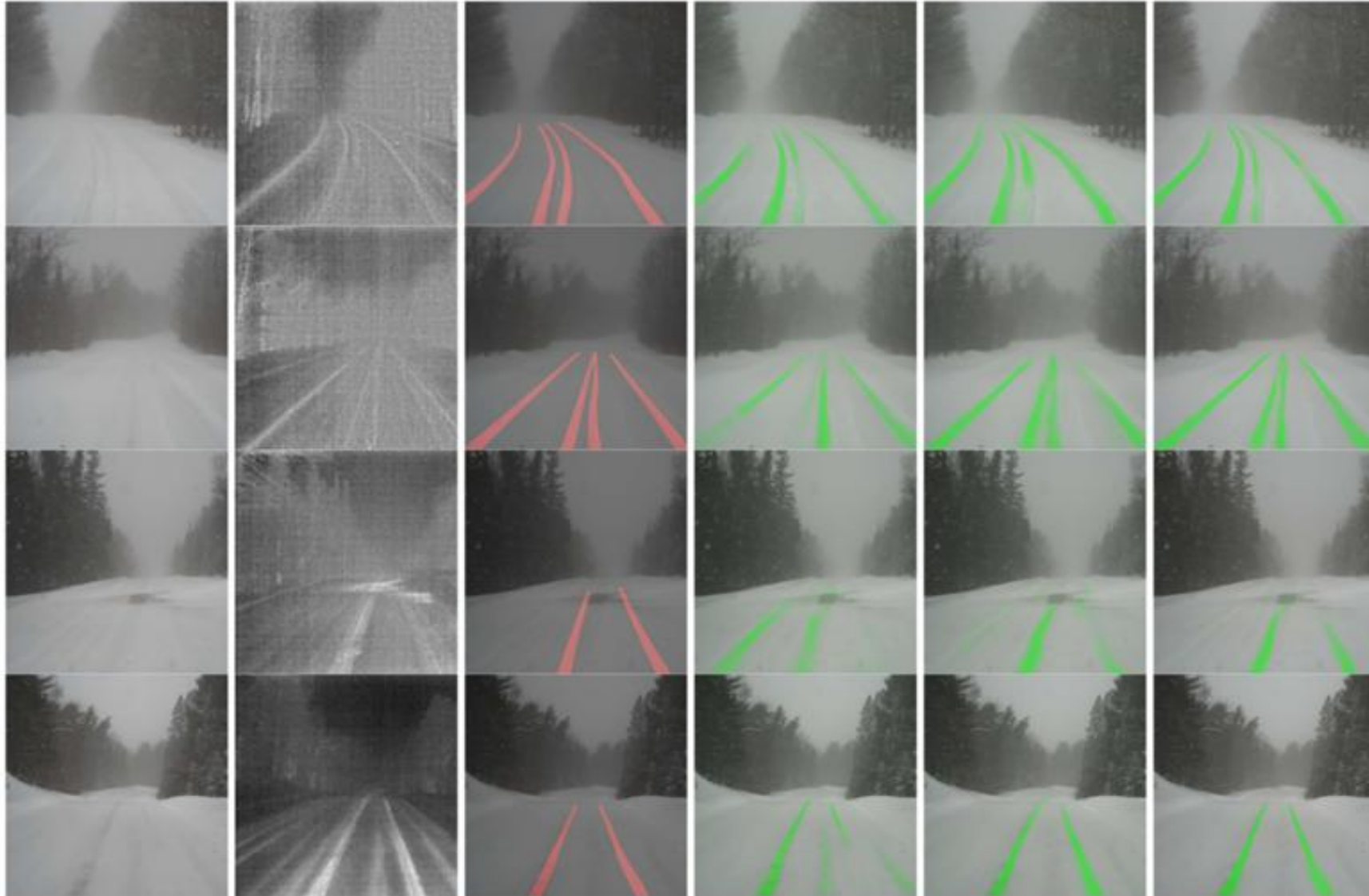


Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion.

Yiming Yang & Jeremy P. Bos



Existing Research



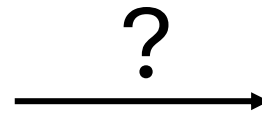
Image-processing [Goberville et al, 2022]



Obvious Wheel tracks



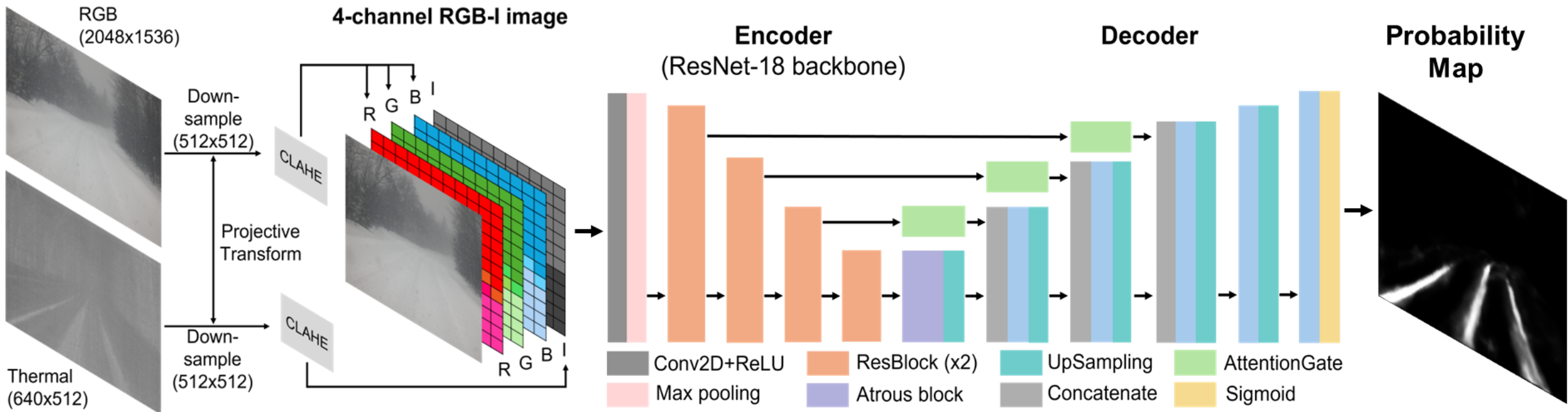
Lidar-based [Fuentes et al., 2021]



Challenge



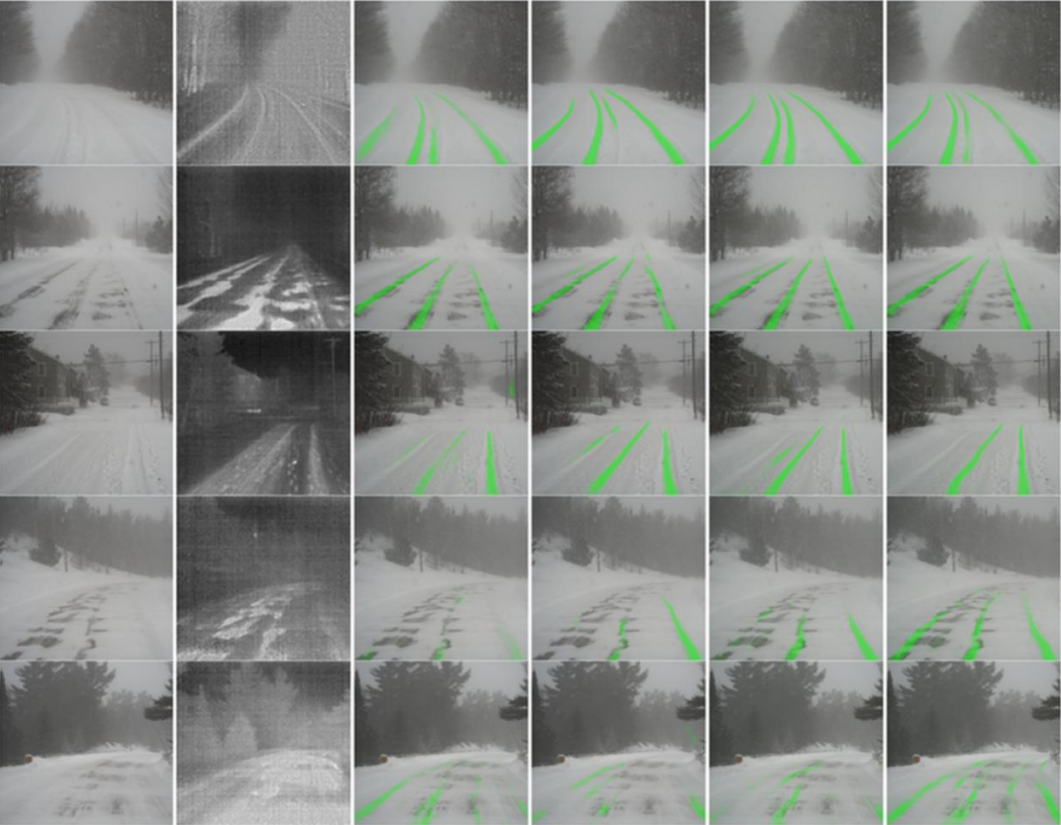
Proposed Approach



Results

Overall Comparison

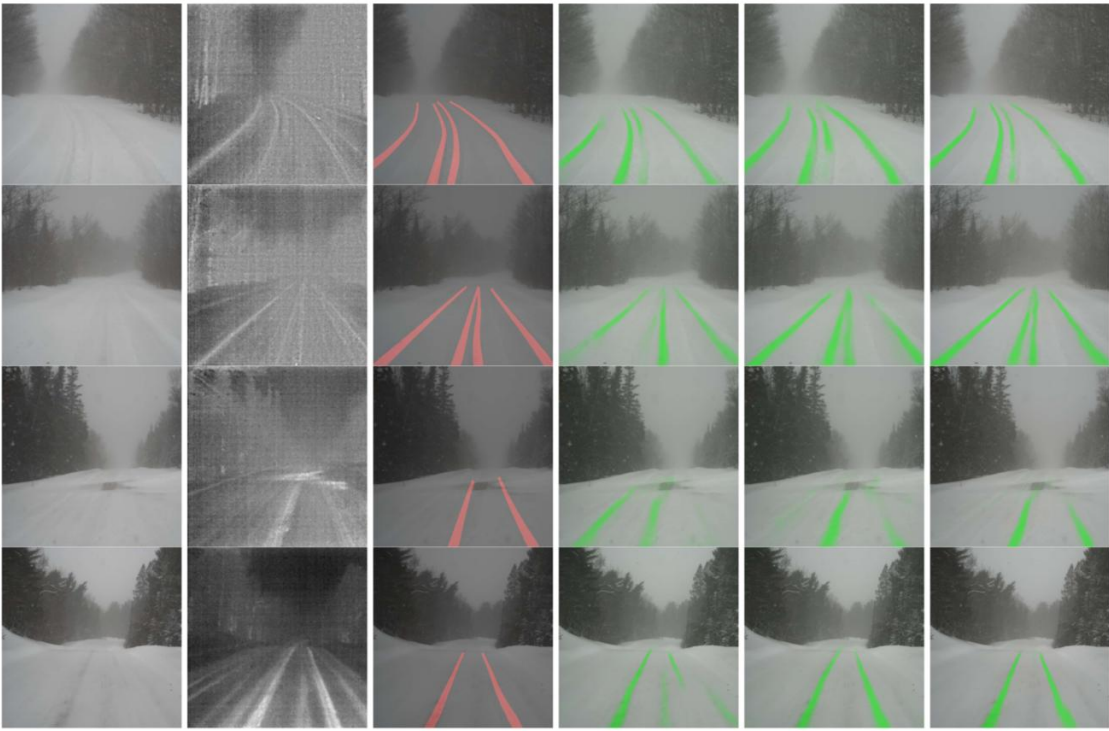
Model	AUROC	MaxF	AP	PRE	REC	FPR	FNR
UNet	0.7068	0.5197	0.6079	0.7691	0.4192	0.0056	0.5808
PSPNet	0.7532	0.6102	0.6666	0.7978	0.5122	0.0057	0.4878
DeepLabv3+	0.7446	0.6002	0.6629	0.8074	0.4943	0.0052	0.5057
ResNet18	0.8462	0.7202	0.7337	0.7499	0.7034	0.0109	0.2966



RGB Thermal Unet+a2 PSPNet+a1 Deeplabv3+a1 Ours

Ablation Study

Model	AUROC	MaxF	AP	PRE	REC	FPR	FNR
UNet+a2	0.7843	0.6529	0.6941	0.7929	0.5753	0.0068	0.4247
PSPNet+a1	0.8476	0.7186	0.7337	0.7471	0.7064	0.0112	0.2936
DeepLabv3+a1	0.8589	0.7460	0.7584	0.7757	0.7281	0.0102	0.2719
Ours-rgb	0.6025	0.3083	0.5164	0.6901	0.2073	0.0024	0.7927
Ours-i	0.7687	0.5803	0.6127	0.6525	0.5510	0.0137	0.4490
Ours-rgb-i	0.8744	0.7579	0.7675	0.7633	0.7601	0.0114	0.2399



RGB Thermal GT Ours-rgb Ours-i Ours-rgb-i

Conclusion

- Thermal image improves wheel track detection
 - Reducing the False Negative Rate by ~70% compared to RGB
 - Maintaining a low False Positive Rate (0.014)

Future Work

- More diverse and unseen terrains
 - e.g., slush, ice patches, mud pits, flooded roads
- Explore generative AI for better uncertainty handling

Thank you! See my poster at Pod D13!

Contact Information

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ThermalTrack Dataset:

<https://digitalcommons.mtu.edu/thermaltrack/>



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THERMALTRACK